

Xi Zhang

AI Engineer · LLM Systems, RAG & Full-Stack AI Products

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SUMMARY

AI Engineer with 5+ years building production LLM and data systems at a financial advisory firm. Designs agentic LLM orchestration, retrieval-augmented generation (RAG), and NL-to-SQL / graph-query pipelines on Azure, and ships them end to end, from Python services and data pipelines to streaming React front ends. Independently built and launched five AI products on Gemini and Claude. MS in Data Science with an economics background, bringing financial and valuation domain depth to AI work.

TECHNICAL SKILLS

LLM & AI: Azure OpenAI, Claude API, Gemini API, LangChain, LangGraph, RAG, tool / function calling, agent orchestration, prompt engineering, semantic search, FAISS, Azure AI Search, Azure Form Recognizer

ML & Data Science: Python, SQL, R, scikit-learn, PyTorch, TensorFlow, time series, causal inference, Pandas, NumPy

Backend & Data: Flask, Quart, Node.js / Express, REST, Server-Sent Events, SQLAlchemy, Pydantic, PostgreSQL / Supabase, SQL Server, Cosmos DB (Gremlin), Redis, MongoDB

Frontend: TypeScript, React 19, Next.js, Vite, Tailwind CSS, D3.js, Recharts, React Three Fiber

Cloud & DevOps: Azure Functions, Blob Storage, Azure AD OAuth / OIDC, Docker, Azure DevOps CI/CD, GitHub Actions, Google Cloud Run, Vercel

EXPERIENCE

Stout

Irvine, CA

Associate, Digital & Data Analyst

May 2021 – Present

Design and build production LLM applications, data pipelines, and full-stack tools used by the firm's advisory teams.

- **Enterprise AI analytics platform:** Architected LLM orchestration that grounds natural-language questions in the database schema to generate SQL and Cosmos DB Gremlin graph queries, running up to 5 governed tool calls per interaction.
- **Document intelligence & RAG:** Engineered an adaptive RAG platform with dynamic query routing and a 95K-token context budget; built LangGraph map-reduce summarization that processes long documents in 25K-token chunks.
- **Scenario optimization platform:** Built a full-stack application (Flask, JavaScript, SQL Server) with OpenAI function calling and RAG that automates a 9-step manual analysis pipeline, including an 11-module scenario configuration workflow.
- **Streaming LLM interfaces:** Delivered real-time chat using Server-Sent Events with incremental Markdown rendering, and hardened output with DOMPurify sanitization against XSS.
- **Data pipelines & infrastructure:** Built Azure Functions ingestion handling 50+ files per project with in-browser PDF review; scaled the concurrent service to 100 Waitress threads behind Azure AD OAuth / OpenID Connect SSO.
- **ML valuation models:** Automated a digital-asset valuation workflow using multi-model pricing (Random Forest, SVR, Ridge / Lasso) to produce scenario-based financial models.
- **AI observability:** Built a Power BI prompt-intelligence dashboard tracking engagement and funnel conversion across AI tools; maintained CI/CD pipelines in Azure DevOps.

SELECTED PROJECTS — independently built and deployed, live at xzhangfox.com

Flux Career — AI Career Intelligence Platform React 19, TypeScript, Express, Gemini, Supabase, D3.js

- LLM pipeline parses PDF resumes into structured profiles, scores them against a live job description, and rewrites the resume and cover letter; also screens postings for visa-sponsorship restrictions.
- Multi-model routing with automatic fallback cut AI response latency by 40%.

Flux Path — Astrology & Compatibility Engine Next.js, React Three Fiber, Claude API, Gemini, Zustand

- Computes Bazi and Ziwei charts deterministically from real astronomical positions, then grounds an LLM assistant in the computed chart, so the model explains results instead of inventing them.
- Six reading systems on a real-time 3D star chart, plus a two-player drawing game that feeds a compatibility reading.

Flux Finance & Flux Nutrition — Multimodal AI Apps React 19, Gemini Vision / Flash, Supabase, Recharts

- Vision models turn a receipt photo into itemized ledger entries and a meal photo into per-item bounding boxes with gram-level macro estimates; AI coaching is grounded in the user's live data and TDEE models.
- Offline-first state sync with Supabase; 95+ Lighthouse performance score.

AI Bubble Monitor — Automated Market Risk Indicator Next.js, Yahoo Finance API, GitHub Actions

- Daily composite score from 7 percentile-ranked indicators across the Dalio, Shiller, and Minsky frameworks, computed with zero look-ahead bias in a fully automated pipeline.

EDUCATION

The George Washington University — M.S., Data Science

Dec 2020

University of California, San Diego — B.S., Economics

Dec 2017